

ELECTRONIC EVIDENCE CERTIFICATE DRAFT
Certificate Draft — Pending Review and Signature

Draft Ref: DEMO-2026-001 | Generated: 2026-09-06 04:47:54 UTC | Tool: Phoenix 0.1.0

These are observed facts produced by an automated pipeline. They are input for a certificate draft under BSA 2023 section 63; they are not a certificate, are unsigned, and require review and signature by the person responsible for the device or its operation.

PART A — DEVICE IDENTITY AND CUSTODY

Case ID	CASE-DEMO-001
Evidence ID	ev-acq-711f8ebe-32c2-4072-bbfc-b0b10da8df93
Source Device	simulated Hikvision-shaped DVR disk, 32 MiB, seed 2026
Source Path	D:\sih\Phoenix_37\hardware\acquisition_rig\out\dvr_image.img
Declared Vendor	Hikvision
Vendor Support Status	Native parser, tested against synthetic and reference samples (validated prototype)
Detection Confidence	95%
Detection Rationale	• Hikvision: hikvision_master_sector@0x210, hikvision_hikbtree@0x400 (unexpected offset); summed weight 0.95 • NAL scan: 7/32 windows contain valid Annex-B start codes (density 0.22), codec guess h264 (h264 votes 48, h265 votes 0) • decision: Hikvision (VALIDATED), confidence 0.95
Acquisition Operator	op-amritansh
Acquisition ID	acq-711f8ebe-32c2-4072-bbfc-b0b10da8df93
Acquisition Started (UTC)	2026-09-05 06:23:03
Acquisition Finished (UTC)	2026-09-05 06:23:03
Acquisition Status	COMPLETED
Custody Notes	[none]
Read-Only Acquisition	Yes
Method	The source was opened read-only and copied byte for byte into an image file. SHA-256 and MD5 were computed on the plaintext stream while it was read, before any other processing. The written image was re-read and re-hashed to confirm the copy. Recordings were then located by their bitstream structure, not by any index on the source, so recordings deleted from the recorder's index are recovered along with the rest.
Adapter Used	backend.adapters.generic_carver.GenericCarverAdapter
Generic Fallback?	Yes — backend.adapters.hikvision.HikvisionAdapter unavailable (vendor adapter is a stub (Sibam: WFS signature detection pending)); using generic carver
Pipeline Stages	intake → detection → recovery → encryption
Platform	Windows-11-10.0.26200-SP0 (Python 3.12.5)

PART B — TECHNICAL INTEGRITY AND RECOVERY

Source Image SHA-256	768e8d4b7d8def560803a6350f07f9c8240cb02351df88af2e1f8c6fa5fa556b
Source Image MD5	38d700ab355afcd069ee6694ed73ff44
Image Size	32.0 MiB (33,554,432 bytes)
Verification Matched	YES — copy verified against source hash

Integrity Statement	The image on disk was re-hashed after writing and matched the hash computed while reading the source, so the copy is identical to what was read. Each recovered fragment was hashed from the image before encryption; any later change to a fragment or to the image is detectable by recomputing these digests.
Recovery Hash	01b574422b990b058094355a771d0d7d120d4e5a641d265b19109149cc618725
Encrypted Artifacts	8 fragment(s) + full image
Recovered Fragments	8
Probable Channels	3
Estimated Footage	56.44 seconds

Hash Lineage — all pipeline stages (plaintext before encryption):

Stage	Algorithm	Hex Digest	Timestamp (UTC)
intake	SHA-256	768e8d4b7d8def560803a6350f07f9c8240cb02351df88af2e1f8c6fa5202550b	2026-09-05 06:23:03
intake	MD5	38d700ab355afcd069ee6694ed73ff44	2026-09-05 06:23:03
intake_verify	SHA-256	768e8d4b7d8def560803a6350f07f9c8240cb02351df88af2e1f8c6fa5202550b	2026-09-05 06:23:03
pre_encryption/fragment_0000	SHA-256	c59cc0052616c6756b744c676c8ab8f602d075521229b396a64379078d893f91	2026-09-05 06:23:04
pre_encryption/fragment_0001	SHA-256	65b9b467decbec9496cd643e503cad601e51c86292c7466321a9d74f285d5194	2026-09-05 06:23:04
pre_encryption/fragment_0002	SHA-256	ce274967a87bc92e22eded5074795a0b60c9dce5d291222e98bf75d2f4fd54bc	2026-09-05 06:23:04
pre_encryption/fragment_0003	SHA-256	0fc569d439301f93c6a76c9180373241859470caadca01983878c64dbd1600c0	2026-09-05 06:23:04
pre_encryption/fragment_0004	SHA-256	dc03aa6681e4a2bf843de39fa6d8beee086b2dbec2230736445928862b783645	2026-09-05 06:23:04
pre_encryption/fragment_0005	SHA-256	473a9686267681976ec7db333c90c6f3b1196f441a2024a5648c9b920e28a78f	2026-09-05 06:23:04
pre_encryption/fragment_0006	SHA-256	bafef4a9c7214e1678676296328aaefb47a0a5dbea631bc8c247998770d0f2d6	2026-09-05 06:23:04
pre_encryption/fragment_0007	SHA-256	bb0a10a9cd0a6b3e3e5583f78ad6fe0cbeb6427c74197a7ea296cd3194975813	2026-09-05 06:23:04

Channel Summary:

Frame Rate	Rationale
25.0 fps	4 fragment(s) share SPS cffa6807e5123e7d; resolution 1280x720; declared 25 fps; this is a probable channel, not a vendor channel number: two cameras configured id
12.5 fps	2 fragment(s) share SPS 86e1eee4a5e549d2; resolution 704x576; declared 12.5 fps; this is a probable channel, not a vendor channel number: two cameras configured
[not declared]	2 fragment(s) share SPS 25d554f36cd7e61e; resolution 704x576; declared no frame rate; this is a probable channel, not a vendor channel number: two cameras config

Recovered Fragment Inventory:

#	Fragment ID	Byte Range	Codec	Channel	Est. Dur.	Conf.	SHA-256
1	frag-c59cc0052616c6756b744c676c8ab8f602d075521229b396a64379078d893f91	768e8d4b7d8def560803a6350f07f9c8240cb02351df88af2e1f8c6fa5202550b-1334281	H.264 Baseline (profile_idc 66)	probable-1280x720-25.0025 fps (95%ared)	26.0025	95%	c59cc0052616c6756b744c676c8ab8f602d075521229b396a64379078d893f91
2	frag-65b9b467decbec9496cd643e503cad601e51c86292c7466321a9d74f285d5194	65b9b467decbec9496cd643e503cad601e51c86292c7466321a9d74f285d5194-2390586	H.264 Baseline (profile_idc 66)	probable-1280x720-25.0025 fps (95%ared)	26.0025	95%	65b9b467decbec9496cd643e503cad601e51c86292c7466321a9d74f285d5194
3	frag-ce274967a87bc92e22eded5074795a0b60c9dce5d291222e98bf75d2f4fd54bc	ce274967a87bc92e22eded5074795a0b60c9dce5d291222e98bf75d2f4fd54bc-3433631	H.264 Baseline (profile_idc 66)	probable-1280x720-25.0025 fps (95%ared)	26.0025	95%	ce274967a87bc92e22eded5074795a0b60c9dce5d291222e98bf75d2f4fd54bc
4	frag-0fc569d439301f93c6a76c9180373241859470caadca01983878c64dbd1600c0	0fc569d439301f93c6a76c9180373241859470caadca01983878c64dbd1600c0-4486664	H.264 Baseline (profile_idc 66)	probable-1280x720-25.0025 fps (95%ared)	57.6125	90%	0fc569d439301f93c6a76c9180373241859470caadca01983878c64dbd1600c0
5	frag-dc03aa6681e4a2bf843de39fa6d8beee086b2dbec2230736445928862b783645	dc03aa6681e4a2bf843de39fa6d8beee086b2dbec2230736445928862b783645-5355531	H.264 Baseline (profile_idc 66)	probable-1280x720-25.0025 fps (95%ared)	26.0025	95%	dc03aa6681e4a2bf843de39fa6d8beee086b2dbec2230736445928862b783645
6	frag-473a9686267681976ec7db333c90c6f3b1196f441a2024a5648c9b920e28a78f	473a9686267681976ec7db333c90c6f3b1196f441a2024a5648c9b920e28a78f-6583180	H.264 Baseline (profile_idc 66)	probable-1280x720-25.0025 fps (95%ared)	26.0025	95%	473a9686267681976ec7db333c90c6f3b1196f441a2024a5648c9b920e28a78f
7	frag-bafef4a9c7214e1678676296328aaefb47a0a5dbea631bc8c247998770d0f2d6	bafef4a9c7214e1678676296328aaefb47a0a5dbea631bc8c247998770d0f2d6-7633457	H.264 Baseline (profile_idc 66)	probable-1280x720-25.0025 fps (95%ared)	26.0025	95%	bafef4a9c7214e1678676296328aaefb47a0a5dbea631bc8c247998770d0f2d6
8	frag-bb0a10a9cd0a6b3e3e5583f78ad6fe0cbeb6427c74197a7ea296cd3194975813	bb0a10a9cd0a6b3e3e5583f78ad6fe0cbeb6427c74197a7ea296cd3194975813-8508632	H.264 Baseline (profile_idc 66)	probable-1280x720-25.0025 fps (95%ared)	57.6244s	80%	bb0a10a9cd0a6b3e3e5583f78ad6fe0cbeb6427c74197a7ea296cd3194975813

Confidence scores are structural (reconstruction completeness), not content authenticity claims.

LIMITATIONS — VERBATIM FROM PIPELINE

- The source was classified as Hikvision (VALIDATED), but no validated native parser for it was used in this run: backend.adapters.hikvision.HikvisionAdapter unavailable (vendor adapter is a stub (Sibam: WFS signature detection pending)); using generic carver. Recordings were recovered by generic bitstream carving, which reads the video structure itself and not the

recorder's own index.

- No wall-clock time is recoverable from the video bitstream alone. Any date or time in this material must come from the recorder's own index or from the operator, and the recorder's clock should be compared with a reference clock before times are relied on.
- Order is the order fragments appear in the image, which is the recording order only if the recorder had not wrapped around and started overwriting.
- No wall-clock time is available from a bare elementary stream; all times are relative to the start of their own channel.
- 2 of 8 fragment(s) have no declared frame rate; their durations use the assumed rate and would scale linearly with the real one.
- 2 fragment(s) do not end at an end-of-stream marker, so footage may be missing at their tail.
- 2 recording(s) declare no frame rate; their durations assume 25 frames per second and scale directly with the true rate.
- 2 recording(s) do not end at an end-of-stream marker, so material may be missing from their end.
- Confidence scores are structural: they describe how completely a recording was reconstructed, not whether its content is authentic.

LEGAL BASIS

This certificate draft is prepared to support the procedural requirements of Section 63 of the Bharatiya Sakshya Adhiniyam 2023, which provides for a certificate that identifies the device, its operator, the method by which the electronic record was produced, and its integrity as shown by hash values. Processing of the source material for the purposes of this forensic examination falls within the exemptions available to authorised law-enforcement and intelligence-gathering functions under Section 17(1)(c) of the Digital Personal Data Protection Act 2023. That exemption is limited in scope: it applies only to processing by or for state agencies acting within their lawful mandate and does not constitute a general licence to process personal data. This draft does not by itself render the evidence admissible. Admissibility is determined by the court on the basis of the completed, signed certificate and all other relevant circumstances. The tool that generated this draft supports procedural documentation — it does not decide admissibility.

SIGNATURE BLOCK — FOR COMPLETION BY RESPONSIBLE PERSON

<i>Full Name of Certifying Person</i> _____	<i>Official Designation / Rank</i> _____
<i>Responsible For (relation to device)</i> _____	<i>Place</i> _____
<i>Date (DD/MM/YYYY)</i> __ / __ / ____	<i>Signature (with seal if applicable)</i> _____

Left blank deliberately. A certificate under BSA 2023 section 63 is valid only when completed and signed by a person occupying a responsible official position in relation to the device or its management.